

VIETNAM NATIONAL UNIVERSITY, HANOI
INTERNATIONAL SCHOOL



FINAL REPORT
SUBJECT: DATABASE
DATA INFORMATION MANAGEMENT OF VIETJET

Name	Student ID
Lê Hà Trang	22070715
Cao Thị Thu Trang	22070748
Lương Vũ Hoàng Anh	22070673
Vũ Minh Ánh	22070832
Phạm Thị Ngọc Hà	22070706

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1, Introduction

There are many well-known airlines in the aviation industry today, and effectively managing the basic information of each flight is the key to success. Vietjet is one of the world's top 10 safest low-cost airlines in 2022. This further requires a robust database to streamline processes. Therefore, it is necessary to design a database that is suitable for the needs of the airline that allows it to manage customers, ticket information, employees, flight routes, etc effectively.

The database system stores and manages basic information about each flight including information on customers, employees, aircraft, airports, services, flight information, reservations, and maintenance. By developing this data system, Vietjet Air will have the most comprehensive management tool from which to avoid errors in service and management. This is also a competitive advantage for the aviation industry.

Investing in a modern aviation management system with a new relational database is not only a step forward in improving operational efficiency but also a long-term strategy to help the airline maintain a competitive advantage and sustainable development. This system will play a key role in improving customer service, optimizing operations, managing finances, and ensuring safety, thereby creating a solid foundation for future success and growth.

2, Business Description

2.1 Business Narrative

VietJet Air, also known as VietJet Aviation Joint Stock Company, is Vietnam's leading international low-cost carrier. It was the first privately owned airline and became the second private airline to offer domestic services in Vietnam. VietJet Air is one of the five airlines providing domestic civil flights within the country. Established in 2007 with investments from major stakeholders including T&C Group, Sovico Holdings, and HDBank, the airline's visionary founder, Madam Thao, recognized the untapped potential of the low-cost aviation market in Vietnam.

VietJet Air adopted the Low-Cost Carrier (LCC) model combined with high service quality and modern technology for strategic reasons. Firstly, Vietnam and Southeast Asia's aviation markets have a strong demand for affordable airfares, and the LCC model meets this demand by cutting unnecessary services and optimizing costs. Additionally, competitive pricing enables VietJet Air to effectively compete with traditional and other low-cost carriers, expanding its market share. The airline further optimizes operational costs by utilizing a single aircraft type, reducing maintenance and training expenses.

Moreover, VietJet Air enhances revenue through ancillary services such as baggage handling, onboard food and beverages, entertainment, and travel insurance. Investments in modern technology improve operational efficiency and customer experience, complemented by innovative marketing campaigns and attractive promotions that capture public attention. This model not only facilitates VietJet Air's network expansion but also aligns with its long-term

strategic vision to become a leading global airline, ensuring sustainable value for customers and shareholders.

From 2007 to 2011, VietJet Air focused on infrastructure development, fleet acquisition, personnel recruitment and training, establishing management systems, and regulatory compliance to prepare for flight operations. In December 2011, VietJet Air launched its first commercial flight, marking its formal entry into the civil aviation market. With aspirations to provide joy in flying and high-quality service to customers worldwide, the airline aims to expand its international flight network, implement advanced technology, and maintain competitive pricing to meet the increasing demand for air travel.

VietJet Air is committed to offering safe, affordable flights and excellent services, continuously improving service quality, optimizing operational costs, and positively contributing to the community. The airline strives to ensure every passenger enjoys a comfortable and satisfying flying experience. Through detailed analysis and design of operational processes, VietJet Air has chosen suitable software technology to ensure comprehensive and flexible expansion in the future.

2.2 User Story:

VietJet Air User Stories considering the perspectives of passengers, employees, and managers:

2.2.1 Passengers

- **Flight Search & Booking:**

- As a passenger, I want to view detailed flight information (aircraft type, flight schedule including departure and arrival times, number of seat availability, etc.) before booking a ticket to find the best options for my trip.
- As a passenger, I want to book tickets with the option to select additional services easily (food, drink, consign, transfer services, and wifi).

- **Booking Management:**

- As a passenger, I want to view and manage my reservation information (flight information, passenger information, selected services) to promptly make adjustments if incorrect.
- As a passenger, I would like to receive notification of any changes to my flight so that I can make timely adjustments.

2.2.2 Employees

- **Flight Management:**

- As an employee, I want to know and be updated about flight information (schedule, aircraft type, status) to send notifications timely to passengers about schedule changes, etc.
- As an employee, I want to manage all the passenger information for each flight to provide support if they need to change anything about their personal information.
- As an employee, I want to handle additional service requests from passengers (food, drink, consign, transfer services, and wifi).

- **Booking Management:**
 - As an employee, I want to view and manage passenger bookings (Booking date and seat number) to adjust if any problems occur.

2.2.3 Managers

- **Fleet & Schedule Management:**
 - As a manager, I want to control and adjust the flight information system including flight schedule, and status (departure or delay) to make timely changes for unpredictable problems such as bad weather, etc.
 - As a manager, I want to view and update aircraft information including aircraft models, capacity, manufacturer, and maintenance dates to improve service quality and ensure passenger safety.
- **Reporting & Analytics:**
 - As a manager, I want to view reports on ticket sales per flight, per period.
 - As a manager, I want to analyze passenger data (age, gender, booking behavior) to make informed business decisions.

The user story provided to VietJet Air outlines the functional requirements from the perspectives of passengers, employees, and managers. These user stories reflect the essential features and interactions that need to be captured in the database design, which will be represented in an Entity Relationship Diagram (ERD).

For passengers, user stories highlight the need for detailed flight information, the ability to book and manage tickets, and easily select all the additional options they want including food, beverages, and more. drink, etc. This implies that the database must include entities such as Flights, Passengers, Reservations, and Service Types.

From an employee perspective, there is a requirement to manage flight information, passenger data, and service requests. This requires expanding the ERD to include entities such as Employees, Passengers (including passenger information), Flights, and Service Types to reach and help passengers with problems and complete be their responsibility.

Managers need control over aircraft systems and schedule management, along with access to reporting and analytics. Therefore, the ERD must include entities such as Flights, Airports, Airways to ensure that flights have routes that include clear and specific departure and arrival information, departure and arrival times, as well as Aircraft information and Maintenance Reports to improve aircraft quality and ensure safety for both passengers and its employees.

3. ERD and Relation Schema

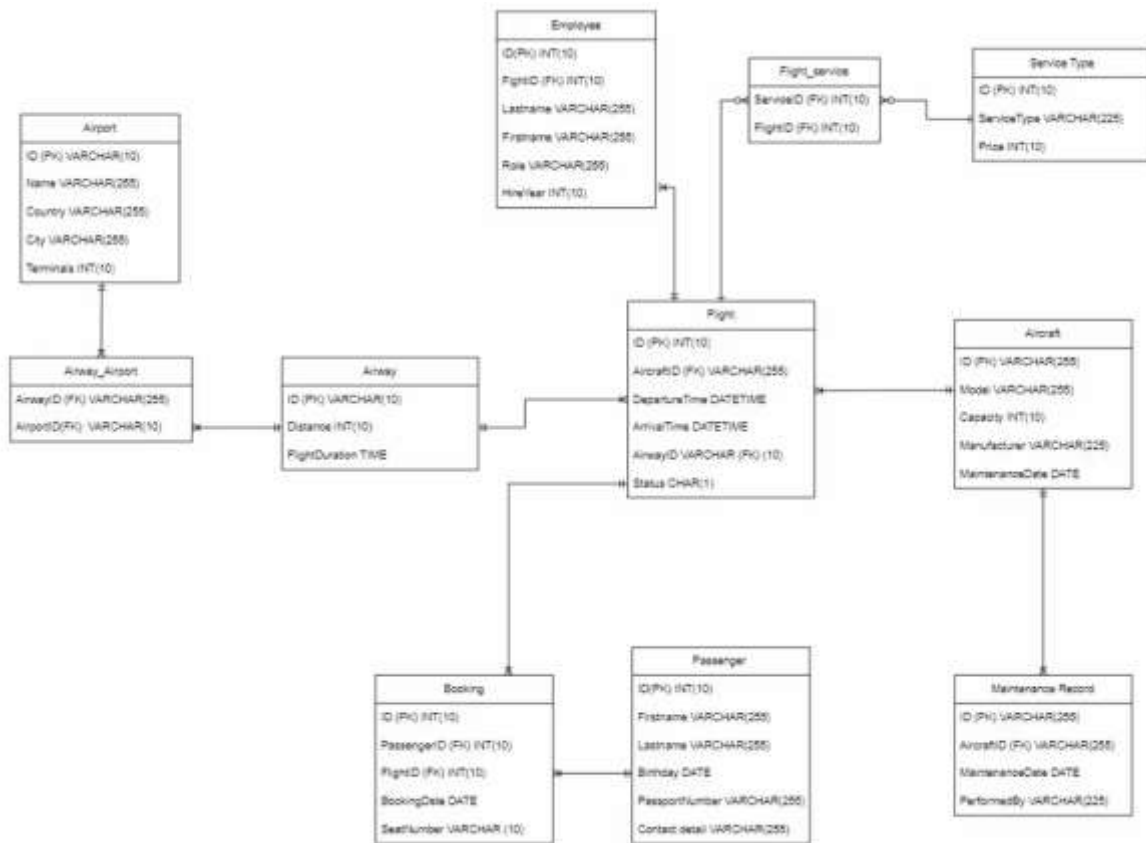
3.1. ERD:

3.1.1 Table detail

1. Flight: Detailed information about flights
 - ID (70): Unique identity for each flight.
 - AircraftID (5): Identify for the aircraft operating the flight
 - DepartureTime: Scheduled departure time.

- ArrivalTime: Scheduled arrival time.
 - AirwayID
 - Status: Current status of the flight (e.g., On-time, Delayed, Cancelled).
2. Aircraft: Detailed information about an aircraft
 - ID: Unique identifier for each aircraft.
 - Model: Model of the aircraft (e.g., A320, A321).
 - Capacity: Number of passengers the aircraft can carry.
 - Manufacturer: The company that manufactured the aircraft.
 - MaintenanceDate: The date of the last maintenance check.
 3. Employees: Airline employee information
 - ID: Unique identifier for each employee.
 - FlightID: Determine the flight that the employee is working on at that time
 - FirstName: Employee's first name.
 - LastName: Employee's last name.
 - Role: Job role (e.g., Pilot, Flight Attendant, Ground Crew).
 - HireYear: The year the employee was hired.
 4. Passengers: Passengers information
 - ID: Unique identifier for each passenger.
 - FirstName: Passenger's first name.
 - LastName: Passenger's last name.
 - Birthday: Passenger's date of birth.
 - PassportNumber: Passport number (for international flights).
 - ContactDetails: Contact information (phone, email).
 5. Bookings: Passenger booking ticket information
 - ID: Unique identifier for each booking.
 - PassengerID: Identifier for the passenger making the booking.
 - FlightID: Identifier for the flight booked.
 - BookingDate: Date the booking was made.
 - SeatNumber: Assigned seat number.
 - PaymentStatus: payment status (e.g., Paid, Pending, Cancelled).
 6. Airports: Information about the airport where the plane will take off or land.
 - ID: Unique identifier for each airport.
 - Name: Name of the airport.
 - City: The city where the airport is located.
 - Country: The country where the airport is located.
 - Terminals: Number of terminals at the airport.
 7. Airway: Information about the aircraft's flight path
 - ID: Unique identifier for each airway.
 - DepartureID: Identifier for the departure airport.
 - ArrivalID: Identifier for the arrival airport.
 - Distance: Distance of the airway.

- FlightDuration: Average duration of the flight on this route.
 - 8. Airway_Airport: Sub table to implement a many-to-many relationship between Airway and Airport.
 - AirwayID: Unique identifier for each airway
 - ArrivalID: Unique identifier for arrival airport
 - DepartureID: Unique identifier for departure airport
 - 9. Service Type: Information on services provided for 1 flight
 - ServiceID: Unique identifier for each service.
 - ServiceType: Type of service (e.g., In-flight meal, Extra baggage).
 - Price: Price of the service.
 - 10. Flight_service: Sub table to implement a many-to-many relationship between services and flights.
 - ServiceID: Unique identifier for each service
 - FlightID: Unique identity for each flight
 - 11. Maintenance Records: Information on aircraft maintenance
 - MaintenanceID: Unique identifier for each maintenance record.
 - AircraftID: Identifier for the aircraft undergoing maintenance.
 - MaintenanceDate: Date of the maintenance.
 - PerformedBy: Identifier for the employee or team that performed the maintenance
- Relationship:
- Flights - Aircraft: One to many: A flight can only be made by 1 aircraft, a plane can make many flights.
 - Aircraft-Airport: Many to many: Each aircraft can have multiple airports, and each airport can have multiple aircraft.
 - Flights - Passengers: One to many: A flight has many customers, but 1 customer can only be present on 1 flight
 - Flights - Employees: One to many: A flight will have many service employees, but 1 employee can only work on 1 flight at the same time.
 - Bookings - Passengers: One to many: Each booking is made by one passenger, but each passenger can have many bookings.
 - Bookings - Flights: One to many: Each booking is for one flight, but each flight can have multiple bookings.
 - Routes - Airports: One to many: Each route connects two airports.
 - Services - Flights: Many to many: Each service can be offered on multiple flights, and each flight can offer multiple services.
 - Maintenance Records-Aircraft: One to many: Each maintenance record is associated with one aircraft, but each aircraft can have multiple maintenance records.



3.2. Relation Schema

Table name: Flight

Attribute	Data type	Primary	Foreign	Constraint
ID	INT(10)	✓		NOT NULL
AircraftID	VARCHAR(255)		Aircraft (ID)	NOT NULL
DepartureTime	DATETIME			
ArrivalTime	DATETIME			
AirwayID	VARCHAR(225)		Airway(ID)	NOT NULL
Status	CHAR(1)			STATUS IN('O', 'D', 'C')

Table name: Aircraft

Attribute	Data type	Primary	Foreign	Constraint
ID	VARCHAR (255)	✓		NOT NULL
Model	VARCHAR(255)			NOT NULL
Capacity	INT(10)			NOT NULL
Manufacturer	VARCHAR(255)			NOT NULL
Maintenance Date	DATE			

Table name: Employee

Attribute	Data type	Primary	Foreign	Constraint
ID	INT(10)	✓		NOT NULL
FightID	INT(10)		Fight(ID)	NOT NULL
Lastname	VARCHAR(255)			NOT NULL
Firstname	VARCHAR(255)			NOT NULL
Role	VARCHAR(255)			NOT NULL
HireYear	INT(10)			NOT NULL

Table name: Passenger

Attribute	Data type	Primary	Foreign	Constraint
ID	INT(10)	✓		NOT NULL
Firstname	VARCHAR(255)			NOT NULL
Lastname	VARCHAR(255)			

Birthday	DATE			
PassportNumber	VARCHAR(255)			
Contact detail	VARCHAR(255)			NOT NULL

Table name: Booking

Attribute	Data type	Primary	Foreign	Constraint
ID	INT(10)	✓		NOT NULL
PassengerID	INT(10)		Passenger (PassengerID)	NOT NULL
FightID	INT(10)		Flight(FightID)	NOT NULL
BookingDate	DATE			
SeatNumber	VARCHAR(10)			NOT NULL

Table name: Airport

Attribute	Data type	Primary	Foreign	Constraint
ID	VARCHAR(10)	✓		NOT NULL
Country	VARCHAR(255)			NOT NULL
Terminals	INT(10)			NOT NULL
Name	VARCHAR(255)			NOT NULL
City	VARCHAR(255)			NOT NULL

Table name: Airway

Attribute	Data type	Primary	Foreign	Constraint
-----------	-----------	---------	---------	------------

ID	Varchar(255)	✓		NOT NULL
Distance	INT(10)			NOT NULL
FlightDuration	TIME			

Table name: Service_Type

Attribute	Data type	Primary	Foreign	Constraint
ID	INT(10)	✓		NOT NULL
ServiceType	VARCHAR(255)			NOT NULL
Price	INT(10)			NOT NULL

Table name: Maintenance_Record

Attribute	Data type	Primary	Foreign	Constraint
ID	VARCHAR(255)	✓		NOT NULL
AircraftID	VARCHAR(255)		Aircraft (ID)	NOT NULL
MaintenanceDate	DATE			
PerformedBy	VARCHAR(255)			NOT NULL

Table name: Fight_service

Attribute	Data type	Primary	Foreign	Constraint
ServiceID	INT(10)		Service (ID)	NOT NULL
FightID	INT(10)		Fight (ID)	NOT NULL

Table name: Airway_Airport:

Attribute	Data type	Primary	Foreign	Constraint
AirwayID	VARCHAR(225)		Airway (ID)	NOT NULL
AirportID	VARCHAR(10)		Airport (ID)	NOT NULL

4. SQL statement:

4.1. Create table:

Create Flight table:

```
CREATE TABLE Flight (
    ID INT NOT NULL PRIMARY KEY,
    Departure_time DATETIME,
    Arrival_time DATETIME,
    AircraftID varchar(255) NOT NULL,
    AirwayID VARCHAR(225) NOT NULL,
    Status_FL CHAR(1) CHECK (Status_FL IN ('O', 'D', 'C')),
    CONSTRAINT FK_FlightAircraft FOREIGN KEY (AircraftID) REFERENCES Aircraft(ID),
    CONSTRAINT FK_FlightAirway FOREIGN KEY (AirwayID) REFERENCES Airway(ID)
);
```



Create Aircraft table:

```
create table Aircraft(
ID int not null primary key,
Model varchar(255) not null,
Capacity int not null,
```

Manufacturer varchar(255) not null,
 Maintenance_date date,
);

```
--Create table Aircraft;
ID varchar(255) not null primary key,
Model varchar(255) not null,
Capacity int not null,
Manufacturer varchar(255) not null,
Maintenance_date date,
);
```

100%
 Messages
 Command completed successfully.
 Completion time: 2024-04-17 13:16:10.00: 20240417:00

Create Employee table

Create table Employee(
 ID int not null primary key,
 FlightID int not null foreign key references Flight(ID),
 Lastname varchar(255) not null,
 Firstname varchar(255) not null,
 Role_employee varchar(255) not null,
 HireYear int not null
);

```
--Create table Employee;
ID int not null primary key,
FlightID int not null foreign key references Flight(ID),
Lastname varchar(255) not null,
Firstname varchar(255) not null,
Role_employee varchar(255) not null,
HireYear int not null,
);

--Create table Passenger;
ID int not null primary key,
```

100%
 Messages
 Command completed successfully.
 Completion time: 2024-04-17 13:16:10.00: 20240417:00

Create Passenger table

Create table Passenger(
 ID int not null primary key,
 Firstname varchar(255) not null,
 Lastname varchar(255) not null,
 Birthday date,
 PassportNumber varchar(255),
 Contact_detail varchar(255) not null
);

```
--Create table Passenger;
ID int not null primary key,
Firstname varchar(255) not null,
Lastname varchar(255),
Birthday date,
PassportNumber varchar(255),
Contact_detail varchar(255) not null,
);

--Create table Booking;
ID int not null primary key,
```

100%
 Messages
 Command completed successfully.
 Completion time: 2024-04-17 13:16:10.00: 20240417:00

Create Booking table

Create table Booking(
ID int not null primary key,
PassengerID int not null foreign key references Passenger (ID),
FlightID int not null foreign key references Flight (ID),
BookingDate date,
Seatnumber varchar(10) not null
);

```
--Create table Booking(  
ID int not null primary key,  
PassengerID int not null foreign key references Passenger (ID),  
FlightID int not null foreign key references Flight (ID),  
BookingDate date,  
Seatnumber varchar(10) not null  
);  
  
--Insert into Service_type (ID, Service_type, Price)  
Values (1, 'Food', 10),  
(2, 'Drink', 5);  
100%  
@ Messages  
Database completed successfully.  
Completion time: 2024-04-17T13:48:48.48814977:00
```

Create Airport table:

Create table Airport(
ID varchar(10) not null primary key,
Country varchar(255) not null,
Terminals INT not null,
Nameairport varchar(255) not null,
City varchar(255) not null
);

```
--Create table Airport(  
ID varchar(10) not null primary key,  
Country varchar(255) not null,  
Terminals INT not null,  
Nameairport varchar(255) not null,  
City varchar(255) not null  
);  
  
--Create table Airway(  
ID varchar(225) not null primary key,  
Distance INT not null,  
Flight_duration time not null,  
);  
100%  
@ Messages  
Database completed successfully.  
Completion time: 2024-04-17T13:48:48.48814977:00
```

Create Airway table:

Create table Airway(
ID varchar(225) not null primary key,
Distance INT not null,
Flight_duration time not null,
);

```
--Create table Airway(  
ID varchar(225) not null primary key,  
Distance INT not null,  
Flight_duration time not null,  
);  
  
--Create table Airway_Airport(  
AirwayID varchar(225) not null foreign key references Airway(ID),  
AirportID varchar(10) not null foreign key references Airport(ID),  
);  
100%  
@ Messages  
Database completed successfully.  
Completion time: 2024-04-17T13:48:48.48814977:00
```

Create Airway_Airport table:

Create table Airway_Airport(
AirwayID varchar(225) not null foreign key references Airway(ID),
AirportID varchar(10) not null foreign key references Airport(ID),
);

A screenshot of a SQL IDE showing the successful execution of a CREATE TABLE statement for 'Airway_Airport'. The SQL code is:

```
CREATE TABLE Airway_Airport(  
  AirwayID varchar(225) not null foreign key references Airway(ID),  
  AirportID varchar(10) not null foreign key references Airport(ID),  
);
```

 The status bar at the bottom indicates 'Command completed successfully' and 'Completion time: 2024-04-07T13:05:46.887889+07:00'.

Create Service table:

create table Service_type(
ID int not null primary key,
Service_type varchar(255),
Price INT not null
);

A screenshot of a SQL IDE showing the successful execution of a CREATE TABLE statement for 'Service_type'. The SQL code is:

```
CREATE TABLE Service_type(  
  ID int not null primary key,  
  Service_type varchar(255),  
  Price INT not null,  
);
```

 The status bar at the bottom indicates 'Command completed successfully' and 'Completion time: 2024-04-07T13:05:46.887889+07:00'.

Create Maintenance record table:

create table Maintenance_record(
ID varchar(255) not null primary key,
AircraftID varchar(255) not null foreign key references Aircraft(ID),
MaintenanceDate date,
Performedby varchar(255) not null,
);

A screenshot of a SQL IDE showing the successful execution of a CREATE TABLE statement for 'Maintenance_record'. The SQL code is:

```
CREATE TABLE Maintenance_record(  
  ID varchar(255) not null primary key,  
  AircraftID varchar(255) not null foreign key references Aircraft(ID),  
  MaintenanceDate date,  
  Performedby varchar(255) not null,  
);
```

 The status bar at the bottom indicates 'Command completed successfully' and 'Completion time: 2024-04-07T13:05:46.812181+07:00'.

Create Flight_service:

Create table Flight_service(
ServiceID int not null foreign key references Service_type(ID),
FlightID int not null foreign key references Flight(ID)
);


```

--Create table Flight_service(
ServiceID int not null foreign key references Service_type(ID),
FlightID int not null foreign key references Flight(ID)
);

--Create table Employee(
ID int not null primary key
);

100% +
Messages
Completion time: 2024-06-17 11:41:44.8484760:00

```

4.2 Insert values

4.2.1 Flight table

Insert into Flight(ID,AircraftID, Departure_time, Arrival_time, AirwayID, Status_FL) values

(123,1,'14:00','05:40','H-N','O'),
(456,2,'14:30','07:56','S-D','D'),
(789,3,'15:00','08:12','N-C','C'),
(852,4,'16:50','08:45','J-H','D'),
(776,5,'17:52','10:50','L-S','O')

```

--Insert into Flight(ID,AircraftID, Departure_time, Arrival_time, AirwayID, Status_FL)
values
(123,1,'14:00','05:40','H-N','O'),
(456,2,'14:30','07:56','S-D','D'),
(789,3,'15:00','08:12','N-C','C'),
(852,4,'16:50','08:45','J-H','D'),
(776,5,'17:52','10:50','L-S','O')

--INSERT INTO Booking (ID, PassengerID, FlightID, BookingDate, SeatNumber)
VALUES

100% +
Messages
Completion time: 2024-06-17 11:41:47.4788000:00:00

```

4.2.2 Aircraft table

INSERT INTO Aircraft (ID, Model, Capacity, Manufacturer, Maintenance_date) VALUES

(1,'Boeing 747', 40, 'Boeing', '05-Mar-24'),
(2,'Airbus A320', 50, 'Airbus', '17-Jan-24'),
(3,'Boeing 747 Freighter', 45, 'Boeing', '08-May-24'),
(4, 'Airbus A330-200F', 48, 'Airbus', '26-Apr-24'),
(5, 'Boeing 749', 43, 'Boeing', '28-Feb-24')

```

--INSERT INTO Aircraft (ID, Model, Capacity, Manufacturer, Maintenance_date)
VALUES
(1,'Boeing 747', 40, 'Boeing', '05-Mar-24'),
(2,'Airbus A320', 50, 'Airbus', '17-Jan-24'),
(3,'Boeing 747 Freighter', 45, 'Boeing', '08-May-24'),
(4, 'Airbus A330-200F', 48, 'Airbus', '26-Apr-24'),
(5, 'Boeing 749', 43, 'Boeing', '28-Feb-24')

--Insert into Flight(ID,AircraftID, Departure_time, Arrival_time, AirwayID, Status_FL)
VALUES

100% +
Messages
Completion time: 2024-06-17 11:41:48.4446711:00:00

```

4.2.3 Employee table

insert into Employee(ID, FlightID, Lastname, Firstname, Role_employee, HireYear) values

('0974980528',123,'Bui','Anh Tuan','Pilot',1994),

('0383529885',456,'Vo','Ha Linh','Flight Attendant',1995),
 ('0789305097',789,'Vu','Minh Anh','Ground Crew',2004),
 ('0789456879',852,'Nguyen','Tuan Anh','Purser',2006),
 ('0365789087',776,'Dang','Hoai An','Pilot',2005),
 ('0965459733',852,'Tran','Manh Tuan','Ground Crew',1999),
 ('0789456876',789,'Vuong','Manh Hiep','Flight Attendant',1998),
 ('0984978502',456,'Pham','Thi Hanh','Purser',1987),
 ('0965456722',123,'Vu','Van Tien','Pilot',2001),
 ('0968549528',776,'Do','Manh Dung','first officer',2005),
 ('0879502097',456,'Pham','Thi Nhung','Flight Attendant',2004),
 ('0974193233',776,'Vu','Thi Huong','first officer',1992),
 ('0862200069',123,'Vu','Quang Huy','Pilot',2005),
 ('0367400118',456,'Le','Tran Anh Dung','Ground Crew',1998),
 ('0389260595',789,'Tran','The Truong','first officer',2012),
 ('0862260093',852,'Nhu','Dieu Linh','Pilot',2008),
 ('0975794467',776,'Tran','Ngoc Hung','Purser',2009),
 ('0598485229',852,'Nguyen','Thu Ha','first officer',2010),
 ('0375973372',456,'Do','Thao Nguyen','Flight Attendant',1996),
 ('0931728737',123,'Pham','Thi Chinh','Ground Crew',1998)

```

--Insert into Employee( ID, FlightID, Lastname, Firstname, Role_employee, HireYear)
values
(0974000118, 123, 'Vu', 'Van Tien', 'Pilot', 2001),
(0389260595, 789, 'Tran', 'The Truong', 'first officer', 2012),
(0862260093, 852, 'Nhu', 'Dieu Linh', 'Pilot', 2008),
(0975794467, 776, 'Tran', 'Ngoc Hung', 'Purser', 2009),
(0598485229, 852, 'Nguyen', 'Thu Ha', 'first officer', 2010),
(0375973372, 456, 'Do', 'Thao Nguyen', 'Flight Attendant', 1996),
(0931728737, 123, 'Pham', 'Thi Chinh', 'Ground Crew', 1998)
  
```

4.2.4 Passenger table

Insert into Passenger (ID, Firstname, Lastname, Birthday, PassportNumber, Contact_detail)

Values (1, 'Van Tien' , 'Vu', '01-jan-98', 098723111, 'vuvantien91@yahoo.com'),
 (2, 'Khanh Linh', 'Vu', '02-feb-98', 08663201, 'vklinh123@gmail.com'),
 (3, 'Hoang Nguyen', 'Lai', '01-apr-01', 06892012, 'lhoangnguyen@gmail.com'),
 (4, 'Minh Giam', 'Hoang', '10-jul-04', 07102932, 'hmgiam@yahoo.com'),
 (5, 'Tranh Anh', 'Huynh', '11-jul-03', 07152002, 'huynhtranganh@gmail.com'),
 (6, 'Tien Dat', 'Nguyen', '01-nov-02', 099232011, 'datng123@gmail.com'),
 (7, 'Tuan Anh', 'Trinh Tran', '30-may-99', 06655261, 'tttanh1993@gmail.com'),
 (8, 'Thi Ha', 'Ngo', '13-feb-67', 08710291, 'Nthiha96@yahoo.com'),
 (9, 'Tuan Anh', 'Phung', '15-jul-68', 0901209332, 'ptuananh18@yahoo.com'),
 (10, 'Van Luong', 'Le', '23-sep-78', 01234567, 'lvl9021@gmail.com'),
 (11, 'Steven', 'Hawking', '24-aug-01', 021223, 'Steve@yahoo.com'),
 (12, 'Elizabeth', 'No1', '09-jul-92', 0129590, 'Elizahehe@gmail.com'),
 (13, 'Jack', 'Sparow', '08-may-56', 02134211, 'cuopbien@icloud.com'),
 (14, 'Emi', 'Fukkada', '09-may-01', 069696969, 'JAnnVvv@gmail.com'),

(15, 'Donal', 'Trump', '12-aug-05', 0695443791, 'USAnumber1@gov.com'),
(16, 'Mono', 'Null', '17-sep-95', 0224456677, 'emtraisontung@gmail.com'),
(17, 'Son Tung', 'MTP', '15-jan-95', 081222223, 'skyoiiii123@gmail.com'),
(18, 'Thomas', 'Edison', '20-aug-46', 012445567, 'Thomas1122@gmail.com'),
(19, 'Chong Un', 'Kim', '30-jan-54', 0844296897, 'nuclear@gmail.com'),
(20, 'Thi Thuy Trang', 'Hoang', '24-mar-96', 0833596277, 'hoangtrang161102@gmail.com'),
(21, 'Nhat Minh', 'Doan', '01-may-04', 0897622082, 'minhdoan98@gmail.com'),
(22, 'Minh Duc', 'Le', '28-apr-05', 0867201022, 'lmd71102@gmail.com'),
(23, 'Tuan Phat', 'Huynh', '29-jan-93', 0905072002, 'htuanp45@gmail.com'),
(24, 'Nghia Hiep', 'Do', '10-oct-92', 0907112002, 'dnhiep1234@gmail.com'),
(25, 'Trang Linh', 'Hoang', '21-mar-94', 0901029889, 'htlinh1992@gmail.com'),
(26, 'Linh My', 'Trinh Tran', '19-jan-04', 0367651202, 'tlinhmy99@gmail.com'),
(27, 'Hai Viet', 'Phi', '28-jan-06', 0964230102, 'haivietphi12@gmail.com'),
(28, 'Viet Anh', 'Vu', '27-jun-04', 0964741672, 'Vuvietanhdz@gmail.com'),
(29, 'Kim Lien', 'Ta', '17-sep-95', 0336782012, 'lienta971@gmail.com'),
(30, 'Minh Anh', 'Nguyen', '21-jun-95', 0913219704, 'minhanhxinh@gmail.com'),
(31, 'Hoang Son', 'Tran', '18-oct-94', 0584622082, 'hsontran@gmail.com'),
(32, 'Duc Anh', 'Nguyen', '25-sep-91', 0988722012, 'ducanh711@gmail.com'),
(33, 'Thanh Nhat', 'Hoang', '24-jun-02', 0584622082, 'htn0507@gmail.com'),
(34, 'Bao Chau', 'Vu Ngo', '11-mar-84', 0982329702, 'vungobaochau@gmail.com'),
(35, 'Thanh Duong', 'Dong', '10-may-96', 0868751685, 'Duongngu123@gmail.com'),
(36, 'Hoang Thang', 'Tran Le', '13-jul-84', 0907781819, 'thangtran111@gmail.com'),
(37, 'Cong Thanh', 'Tran', '06-mar-98', 09012345678, 'thanhtrancong@gmail.com'),
(38, 'Tuan Anh', 'Dam', '14-feb-68', 09012345678, 'cosmos6578@gmail.com'),
(39, 'Linh Chi', 'Phung', '23-may-65', 0907781819, 'linhchi113@gmail.com'),
(40, 'Bao Nam', 'Hoang', '23-oct-78', 0868855613, 'hoangnam123@gmail.com'),
(41, 'Phuong Tuan', 'Trinh Tran', '31-aug-98', 098653420, 'j97123@gmail.com'),
(42, 'Thien An', 'Nguyen Hoang', '17-jul-84', 0376590401, 'thienan5tr@gmail.com'),
(43, 'Thien Son', 'Trinh Tran', '05-may-73', 0584308790, 'thienson97@gmail.com'),
(44, 'Dong Nhi', 'Nguyen', '18-aug-03', 098786521, 'dongquaxuantoi@gmail.com'),
(45, 'Anh Tuan', 'Ha', '23-may-92', 0376799245, 'connhoanhkhong@gmail.com'),
(46, 'Luong Anh', 'Nguyen', '14-jun-82', 098367218, 'luonganh07@gmail.com'),
(47, 'Ha Trang', 'Le', '05-sep-01', 0985765439, 'hatrangne@gmail.com'),
(48, 'Thuy Ha', 'Nguyen Thi', '18-jan-00', 0373529776, 'hathaibinh@gmail.com'),
(49, 'Thu Trang', 'Cao Thi', '16-sep-68', 0978568241, 'trangkhi@gmail.com'),
(50, 'Quang Hung', 'Do', '08-Jul-95', 0984980568, 'hungbavi@gmail.com')

```

Insert into Passenger (ID, Firstname, Lastname, Birthday, PassportNumber, Contact_detail)
Values (1, 'Van Tien', 'Vu', '01-Jan-96', 09872333, 'vuocvien93@yahoo.com'),
(2, 'Quang Linh', 'Vu', '02-Feb-98', 09663303, 'skilr012@gmail.com'),
(3, 'Hoang Nguyen', 'Lai', '03-Apr-01', 09892013, 'hoangnguyend@gmail.com'),
(4, 'Minh Giam', 'Hoang', '10-Jul-04', 07182902, 'nglam@yahoo.com'),
(5, 'Frank Lam', 'Nguyen', '11-Jul-01', 07152002, 'franktranganh@gmail.com'),
(6, 'Tien Dai', 'Nguyen', '03-Nov-03', 099222011, 'daieng123@gmail.com'),
(7, 'Tuan Anh', 'Trinh Tron', '10-May-00', 06655263, 'tttrinh1993@gmail.com'),
(8, 'Thu Ha', 'Nguyen', '13-Feb-67', 08716291, 'thiha6@yahoo.com').

```

4.2.5 Booking table

INSERT INTO Booking (ID, PassengerID, FlightID, BookingDate, SeatNumber)
VALUES

(10, 1, 123, '01-Jan-24', 'A1'),
(11, 2, 456, '02-Jan-24', 'A2'),
(12, 3, 789, '03-Jan-24', 'A8'),
(13, 4, 852, '04-Jan-24', 'A9'),
(14, 5, 776, '05-Jan-24', 'A10'),
(15, 6, 123, '06-Jan-24', 'A11'),
(16, 7, 456, '07-Jan-24', 'A12'),
(17, 8, 789, '08-Jan-24', 'A13'),
(18, 9, 852, '09-Jan-24', 'A14'),
(19, 10, 776, '10-Jan-24', 'A15'),
(20, 11, 123, '11-Jan-24', 'A16'),
(21, 12, 456, '12-Jan-24', 'A17'),
(22, 13, 789, '13-Jan-24', 'A18'),
(23, 14, 852, '14-Jan-24', 'A19'),
(24, 15, 776, '15-Jan-24', 'A20'),
(25, 16, 123, '16-Jan-24', 'A21'),
(26, 17, 456, '17-Jan-24', 'A22'),
(27, 18, 789, '18-Jan-24', 'A23'),
(28, 19, 852, '19-Jan-24', 'A24'),
(29, 20, 776, '20-Jan-24', 'B1'),
(30, 21, 123, '21-Jan-24', 'B2'),
(31, 22, 456, '22-Jan-24', 'B3'),
(32, 23, 789, '23-Jan-24', 'B4'),
(33, 24, 852, '24-Jan-24', 'B5'),
(34, 25, 776, '25-Jan-24', 'B6'),
(35, 26, 123, '29-Jan-24', 'B7'),
(36, 27, 456, '12-Jan-24', 'B8'),
(37, 28, 789, '11-Jan-24', 'B9'),
(38, 29, 852, '14-Jan-24', 'B10'),
(39, 30, 776, '15-Jan-24', 'C1'),
(40, 31, 123, '16-Jan-24', 'C2'),
(41, 32, 456, '17-Jan-24', 'C3'),

(42, 33, 789, '18-Jan-24', 'C4'),
 (43, 34, 852, '19-Jan-24', 'C5'),
 (44, 35, 776, '20-Jan-24', 'C6'),
 (45, 36, 123, '21-Jan-24', 'C7'),
 (46, 37, 456, '22-Jan-24', 'C8'),
 (47, 38, 789, '23-Jan-24', 'C9'),
 (48, 39, 852, '24-Jan-24', 'C10'),
 (49, 40, 776, '25-Jan-24', 'C11'),
 (50, 41, 123, '26-Jan-24', 'C12'),
 (51, 42, 456, '27-Jan-24', 'C13'),
 (52, 43, 789, '28-Jan-24', 'C14'),
 (53, 44, 852, '29-Jan-24', 'C15'),
 (54, 45, 776, '30-Jan-24', 'D1'),
 (55, 46, 123, '21-Jan-24', 'D2'),
 (56, 47, 456, '17-Jan-24', 'D3'),
 (57, 48, 789, '16-Jan-24', 'D4'),
 (58, 49, 852, '19-Jan-24', 'D5'),
 (59, 50, 776, '20-Jan-24', 'D6'),
 (60, 40, 123, '21-Jan-24', 'D7'),
 (61, 10, 456, '22-Jan-24', 'D8'),
 (62, 15, 789, '23-Jan-24', 'D9'),
 (63, 12, 852, '24-Jan-24', 'D10'),
 (64, 13, 776, '25-Jan-24', 'D11'),
 (65, 18, 123, '26-Jan-24', 'D12'),
 (66, 20, 456, '27-Jan-24', 'D13'),
 (67, 40, 789, '28-Jan-24', 'D14')



4.2.6 Airport table

Insert into Airport(ID,Country,Nameairport,Terminals,City)

values

('HAN','Việt Nam','Sân bay Quốc tế Nội Bài',4,'Hà Nội'),
 ('SGN','Việt Nam','Sân bay Quốc tế Tân Sơn Nhất',4,'Thành phố Hồ Chí Minh'),
 ('NRT','Nhật Bản','Sân bay Quốc tế Narita',3,'Tokyo'),
 ('JFK','Mỹ','Sân bay Quốc tế John F. Kennedy',5,'New York'),
 ('LHR','Anh','Sân bay Heathrow',4,'London'),

('DAD','Việt Nam','Sân bay Quốc tế Đà Nẵng',5,'Đà Nẵng'),
 ('CXR','Việt Nam','Sân bay Quốc tế Cam Ranh',3,'Khánh Hòa'),
 ('HUI','Việt Nam','Sân bay Phú Bài',2,'Huế'),
 ('VCA','Việt Nam','Sân bay Quốc tế Cần Thơ',3,'Cần Thơ')

```

Insert into Airport (ID, Country, Name, Airport, Terminals, City)
values
('HUI', 'Việt Nam', 'Sân bay Quốc tế Phú Bài', 2, 'Huế'),
('SGN', 'Việt Nam', 'Sân bay Quốc tế Tân Sơn Nhất', 4, 'Thành phố Hồ Chí Minh'),
('NRT', 'Nhật Bản', 'Sân bay Quốc tế Narita', 3, 'Tokyo'),
('JFK', 'Mỹ', 'Sân bay Quốc tế John F. Kennedy', 5, 'New York'),
('LHR', 'Anh', 'Sân bay Heathrow', 4, 'London'),
('DAD', 'Việt Nam', 'Sân bay Quốc tế Đà Nẵng', 5, 'Đà Nẵng'),
('CXR', 'Việt Nam', 'Sân bay Quốc tế Cam Ranh', 3, 'Khánh Hòa'),
('HUI', 'Việt Nam', 'Sân bay Phú Bài', 2, 'Huế')
  
```

4.2.7 Airway table

Insert into Airway (ID, Distance, Flight_duration)

values

('H-N', 2400, '05:00:00'),
 ('S-D', 810, '02:00:00'),
 ('N-C', 2500, '06:00:00'),
 ('J-H', 8500, '10:00:00'),
 ('L-S', 6200, '15:00:00')

```

Insert into Airway (ID, Distance, Flight_duration)
values
('H-N', 2400, '05:00:00'),
('S-D', 810, '02:00:00'),
('N-C', 2500, '06:00:00'),
('J-H', 8500, '10:00:00'),
('L-S', 6200, '15:00:00')
  
```

4.2.8. Airway_Airport table

Insert into Airway_Airport (AirwayID, AirportID)

values

('H-N', 'HAN'),
 ('S-D', 'SGN'),
 ('N-C', 'NRT'),
 ('J-H', 'JFK'),
 ('S-D', 'LHR'),
 ('N-C', 'DAD'),
 ('J-H', 'CXR'),
 ('H-N', 'HUI'),
 ('S-D', 'VCA')

```

--Insert into Airway_Airport(AirwayID, AirportID)
values
('H-M','HNM'),
('S-D','SMT'),
('H-C','HNT'),
('J-H','JKA'),
('S-D','SMT');

```

100 %

Messages

19 rows affected

Completion time: 2024-09-07T10:48:01.881840+07:00

4.2.9. Service_type table

Insert into Service_type (ID, Service_type, Price)

Values (1, 'Food', 10),

(2, 'Drink', 5),

(3, 'Consign', 5),

(4, 'Transfer service', 10),

(5, 'Wifi', 6);

```

--Insert into Service_type (ID, Service_type, Price)
values (1, 'Food', 10),
(2, 'Drink', 5),
(3, 'Consign', 5),
(4, 'Transfer service', 10),
(5, 'Wifi', 6);

--Insert into Passenger (ID, Firstname, Lastname, Birthday, PassportNumber, Contact_detail)
values (1, 'Van Tim', 'Vu', '01-Jan-86', '098723111', 'vuxientian@wefoo.com');

```

100 %

Messages

19 rows affected

Completion time: 2024-09-07T10:48:02.484762+07:00

4.2.10. Maintenance_record table

INSERT INTO Maintenance_record (ID, AircraftID, MaintenanceDate , PerformedBy)

VALUES

('ABZ1482', 1, '14-Feb-24', 'Flyteam'),

('GIK5214', 2, '23-may-23', 'Mtpteam'),

('SPY6541', 3, '12-jan-21', 'Linteam'),

('MAR3125', 4, '20-jul-23', 'Katteam'),

('AWE2385', 5, '8-jun-20', 'Kicteam')

```

--INSERT INTO Maintenance_record ( ID, AircraftID, MaintenanceDate , PerformedBy)
VALUES
('ABZ1482', 1, '14-Feb-24', 'Flyteam'),
('GIK5214', 2, '23-may-23', 'Mtpteam'),
('SPY6541', 3, '12-jan-21', 'Linteam'),
('MAR3125', 4, '20-jul-23', 'Katteam'),
('AWE2385', 5, '8-jun-20', 'Kicteam');

```

100 %

Messages

19 rows affected

Completion time: 2024-09-07T10:48:02.981769+07:00

4.2.11. Flight-Service table

Insert into Flight_service(ServiceID,FlightID)

values

(1,123),

(2,456),

(3,789),

(4,852),

(5,776)

```

Insert into Flight_service(ServiceID,FlightID)
values
(1,123),
(2,456),
(3,789),
(4,852),
(5,776)

```

100 %
 Messages
 (1 row affected)
 Completion time: 2024-09-27T18:09:25.801384+07:00

4.3 Questions:

1. A passenger named Mono wants to double-check his flight ID and seat number and his own passenger ID number on the system to see if they are correct or not.

To list orders (booking) that a customer named Mono has placed, we use the *“Select” statement and “Where” statement*.

Select B.FlightID, B.Seatnumber, P.ID, P.Firstname

from Booking B inner join Passenger P on B.PassengerID = P.ID

where P.Firstname = 'Mono';

```

Select B.FlightID, B.Seatnumber, P.ID, P.Firstname
from Booking B inner join Passenger P on B.PassengerID = P.ID
where P.Firstname = 'Mono';

```

100 %
 Results Messages
 FlightID Seatnumber ID Firstname
 1 123 401 Mono

2. A manager wants to have a list of information about all the airline's customers

To list all customers with specific information, we use *“Select” statements*

SELECT * FROM Passenger

```

SELECT * FROM Passenger;

```

100 %
 Results Messages
 ID Firstname Lastname Birthdate PassportNumber Contact_email
 1 1 Van Tien Hu 1980-01-01 80220171 vovanw@100yolux.com
 2 2 Kiana Linh Hu 1980-01-02 88941201 kiana1210@gmail.com
 3 3 Huong Nguyen Lan 2001-04-01 88300112 huongnguyen@gmail.com
 4 4 Minh Quan Huong 2004-07-10 77520022 minhquan@yahoo.com
 5 5 Frank Anh Huong 2003-07-11 77520032 frankhuong4@gmail.com
 6 6 Tien Dat Nguyen 2002-11-01 86530111 tien1210@gmail.com
 7 7 Tien Anh Tien Tien 1989-05-03 86550181 tien198905@gmail.com
 8 8 Tin Hoa Ngo 1987-02-12 87752081 tinhoa@yahoo.com
 9 9 Tien Anh Phung 1980-07-10 86720032 tiananh1@yahoo.com
 10 10 Van Luong La 1976-02-22 1234567 vanluong1@gmail.com
 11 11 Steven Huong 2001-06-24 11223 Stevenhuong.com
 12 12 Elizabeth Hu 1992-07-09 123028 Elizabeth@gmail.com
 13 13 Jack Sparrow 1980-05-09 2734211 jacksparrow@icloud.com
 14 14 Eric Fukuda 2001-05-09 88909069 ericmwa@gmail.com
 15 15 Daniel Trump 2005-06-12 89443791 danieltrump1@gmail.com

3. A customer service employee wants to be able to quickly look up flight information based on the flight code the customer provides to support customers more effectively.

To meet the requirements of customers, we create a *procedure* to search for flights by flight code:

- Create procedure

CREATE PROCEDURE uspFlightInfor

@ID int

as

begin


```

select *
from flight
Where ID = @ID;
End;

```



- Once created, we tested it with flight code 123:
Execute uspFlightInfo @ID=123



- A flight attendant on flight 123 wants a complete list of information on all passengers on this flight to verify their identities when passengers board the plane and during service.

To list all customers who have booked a flight with a flight code = 123, we use **inner join** statement and **where** statement.

```

select B.FlightID, B.ID, P.ID, P.Firstname, P.Lastname, P.PassportNumber
from Booking B inner join Passenger P on B.PassengerID = P.ID
where B.FlightID=123;

```



- A Vietjet Air flight (number 123) had a technical problem that could not be fixed immediately. The airline decided to postpone the flight and the employees needed to urgently notify all passengers who had booked tickets so they could reschedule their schedules. To contact passengers, the airline employee needs to get their name and contact information from the database.

To Retrieve the names and phone numbers of passengers who have booked tickets on the flight with flight number 123, we use **the 'select' statement**.

```

SELECT p.FirstName, p.LastName, p.Contact_detail
FROM Passenger p
WHERE p.ID IN (
    SELECT b.PassengerID
FROM Booking b

```

WHERE b.FlightID = 123

);

100% Results Messages

ProfileNo	LastName	Contact_email
1	Tran Thi	tran.thi1@yahoo.com
2	Tran Dai	daiky123@gmail.com
3	Binh	binh@yahoo.com
4	Ngoc	ngoc.ngoc@gmail.com
5	Thao My	thao.my123@gmail.com
6	Nguyen Thi	nguyen.thi@gmail.com
7	Tran Thi	tran.thi123@gmail.com
8	Tran Thi	tran.thi123@gmail.com
9	Tran Thi	tran.thi123@gmail.com
10	Tran Thi	tran.thi123@gmail.com
11	Tran Thi	tran.thi123@gmail.com
12	Tran Thi	tran.thi123@gmail.com

- Vietjet Air's technical staff are periodically checking the maintenance history of the entire fleet. They need a detailed report that includes information about each aircraft including ID and model as well as corresponding maintenance-related information to ensure compliance with safety regulations and evaluate the effectiveness of maintenance procedures.**

To retrieve information about the aircraft and their corresponding maintenance information, we use **inner join** statement.

select A.ID, A.Model, M.ID, M.MaintenanceDate, M.Performedby
from Maintenance_record M inner join Aircraft A on A.ID = M.AircraftID

100% Results Messages

ID	Model	ID	MaintenanceDate	Performedby
1	Boeing 747	4021402	2024-12-16	Phuoc
2	Boeing 747	3092109	2023-08-02	Phuoc
3	Boeing 747	3092109	2023-08-02	Phuoc
4	Boeing 747	3092109	2023-08-02	Phuoc
5	Boeing 747	3092109	2023-08-02	Phuoc

- Vietjet Air's manager is analyzing flight performance to optimize resource allocation and future flight planning. They want to know which flights have the highest number of passengers and the earliest departure times so they can focus on promoting and upgrading services for these flights.**

To retrieve the list of flights along with information about the number of passengers, sorted by the number of passengers in descending order and the departure time in ascending order, we use **group by and order by, count** statements.

Select F.ID, F.AircraftID, F.AirwayID, F.Departure_time, Count(B.PassengerID)
number_Passenger

from Flight F inner join Booking B on F.ID = B.FlightID

Group by F.ID, F.AircraftID, F.AirwayID, F.Departure_time

Order by number_Passenger DESC, F.Departure_time ASC;

100% Results Messages

ID	AircraftID	AirwayID	Departure_time	number_Passenger
1	123	1	1000-01-01 14:00:00.000	12
2	408	2	1000-01-01 14:30:00.000	12
3	108	3	1000-01-01 15:00:00.000	12
4	852	4	1000-01-01 18:00:00.000	11
5	778	5	1000-01-01 17:52:00.000	11

8. Vietjet Air's manager is planning to expand its flight network. To make strategic decisions, they need to analyze the current airports that the airline is operating, specifically the number of routes at each airport.

To find airports with at least 1 flight, including detailed information about the airport and the total number of flights, then sort the results in descending order we use **group by, having, order by** statements.

```
Select P.ID, P.Nameairport, P.Country, P.Terminals, count(A.AirwayID)
number_airway
from Airway_Airport A inner join Airport P on P.ID = A.AirportID
Group by P.ID, P.Nameairport, P.Country, P.Terminals having count(A.AirwayID)
>=1
Order by number_airway DESC;
```



ID	Nameairport	Country	Terminals	number_airway
1	SSC Sân bay Quốc tế Cát Bi	VN/Hà Nội	2	1
2	DAD Sân bay Quốc tế Đà Nẵng	VN/Hà Nội	8	1
3	HAN Sân bay Quốc tế Nội Bài	VN/Hà Nội	4	1
4	HFA Sân bay Phú Bài	VN/Hà Nội	2	1
5	PHB Sân bay Quốc tế Phan Thiết	PH/Phước Ninh	5	1
6	LHR Sân bay Heathrow	UK/London	4	1
7	MTT Sân bay Quốc tế Thủ Đức	VN/Hà Nội	2	1
8	SGM Sân bay Quốc tế Tân Sơn Nhất	VN/Hà Nội	4	1
9	KOA Sân bay Quốc tế Chu Lai	VN/Hà Nội	2	1

9. Vietjet Air is having a special promotion for all passengers whose names contain the letter "a". To implement this program, the airline needs a list of all flights with passengers meeting the above conditions, including detailed information about the flight and departure airport.

To Retrieve the list of flights where the first name of the passenger contains the character "a" including detailed information about the flight and the departure airport then sort the results in ascending orders we use **position, order by, and inner join** statement.

```
select B.FlightID, B.PassengerID, P.ID, P.Firstname
from Booking B inner join Passenger P on P.ID = B.PassengerID
where P.Firstname like '%a%'
Group by B.FlightID, B.PassengerID, P.ID, P.Firstname
Order by B.FlightID ASC;
```



FlightID	PassengerID	ID	Firstname
1	1	1	Van Tuan
2	8	8	Tuan Dai
3	18	10	Thomson
4	21	21	Nguyen Minh
5	31	31	Hoang Son
6	36	36	Hoang Thang
7	48	40	Nguyen Nam
8	61	41	Phuong Thuan
9	68	48	Luong Son
10	7	2	Khac Linh
11	7	7	Tuan Dai
12	10	10	Van Luong
13	12	12	Khac Linh
14	20	20	Thao Thuan
15	21	21	Nguyen Nam
16	32	32	Nguyen Anh
17	31	31	Nguyen Thuan

10. Customers want to know the actual flight time of a flight with arrival time = 05:40 và departure time = 14:40 :

To fulfill that requirement, we created a function to calculate the actual flight time by knowing the departure and arrival times.

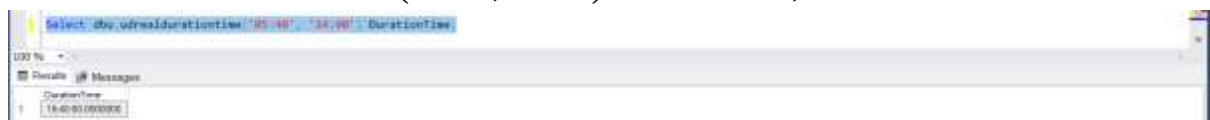
Firstly, we create “udrealdurationtime” function in SQL. In this function, there are 2 variables, arrival time and departure time, which are in the form of time data and return the Duration variable as Time.

```
CREATE FUNCTION udrealdurationtime(  
    @Arrival TIME,  
    @Departure TIME  
)  
RETURNS TIME  
AS  
BEGIN  
    DECLARE @Duration TIME;  
    DECLARE @DurationSeconds INT = DATEDIFF(SECOND, @Departure,  
@Arrival);  
    SET @Duration = CONVERT(TIME, DATEADD(SECOND,  
@DurationSeconds, '00:00:00'));  
    Return @Duration;  
END;
```



Then we use the "udrealdurationtime" function with arrival time = 05:40 and departure time = 14:40. Actual flight time is 15:40

Select dbo.udrealdurationtime('05:40', '14:00') DurationTime;



5. Conclusion:

In summary, developing a powerful and comprehensive database system for VietJet Air is a significant step towards improving operational efficiency and maintaining a competitive advantage in the aviation industry. Detailed database design, including nine main entities such as Flights, Passengers, Booking, Employees, Aircraft, Airports, Airways, Service Types, and Maintenance Records ensures that all critical aspects of airline operations are meticulously managed. Implementing a relational database facilitates the integration of

functions ranging from flight scheduling and passenger management to maintenance tracking and service delivery.

By addressing the needs of passengers, employees, and managers through clearly defined user stories and entity relationships and based on a pre-existing dataset, we leveraged SQL as a tool to design a clear and logical database system that aims to not only improve service quality and operational transparency but also enable informed decision making through reporting and Advanced analytics. This system supports VietJet Air's strategic vision of providing safe, affordable, and high-quality air travel services while optimizing costs and enhancing customer satisfaction to ensure long-term goals can be achieved.